

Review 04 — Axicon / Bessel beams

mcddev Phase 4 (output/04_axicon/, docs/theory/04_axicon.md)

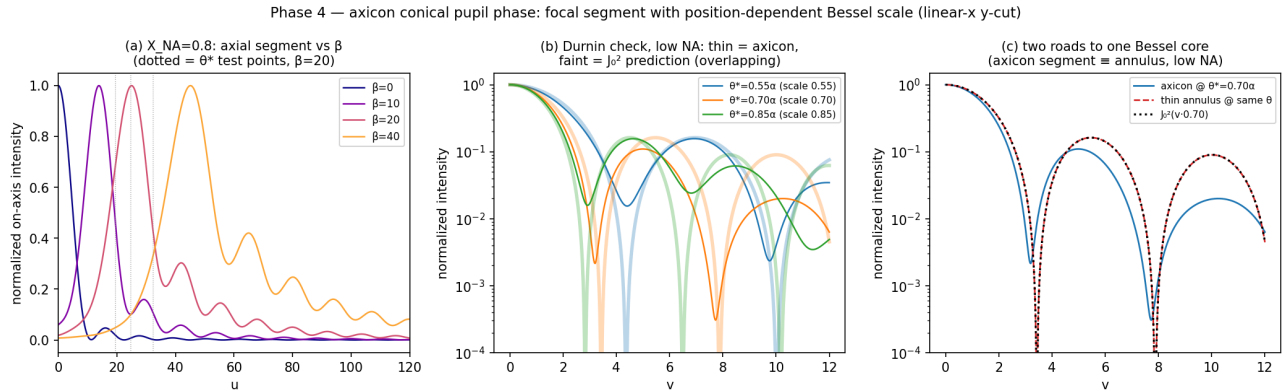
2026-06-13

Construction

Conical pupil phase $A(\theta) = \exp(+j\beta \sin \theta / \sin \alpha)$ (true axicon: phase $\propto$ pupil radius). Stationary phase maps axial position to cone angle, $u(\theta^*) = \beta \sin \alpha \cot \theta^*$, with local transverse profile $J_0^2(v \sin \theta^* / \sin \alpha)$ [Durnin 1987]. Linear- x y-cut.

- $\beta = 0$ regression $\equiv$ plain disk: max diff 0 (exact).
- Focal segment forms at $u > 0$, onset $u \approx \beta \cos \alpha$; bright length 10.9 / 14.2 / 17.9 for $\beta=10/20/40$ (plain focus FWHM 9.9).
- **Durnin checkpoint:** at $\theta^*=0.55/0.70/0.85\alpha$ the transverse profile at the predicted u matches $J_0^2(v \sin \theta^* / \sin \alpha)$ to max dev **0.04–0.08**, at both $X_{NA}=0.13$ and 0.8.
- Axicon local profile $\equiv$ thin annulus at the matched θ — two roads to one Bessel core.

Flag (honest): the 0.04–0.08 J_0^2 deviation is *comparable at low and high NA*, so it is the stationary-phase-approximation error (finite segment + amplitude taper), *not* a vector effect. The clean vector-Bessel probe is the thin annulus (Phase 3: dev 0.163 at $\sin \alpha=0.9$ vs 0.004 low NA). The axicon's DOF gain (~ 1.4 – $1.8\times$) is modest next to the annulus ($\sim 44\times$); its value is the position-tunable Bessel scale.



- (a) axial segment vs β (dotted = θ^* test points); (b) low-NA Durnin check (thin = axicon, faint thick = J_0^2); (c) axicon $\equiv$ annulus $\equiv J_0^2$ at $\theta^*=0.70\alpha$.

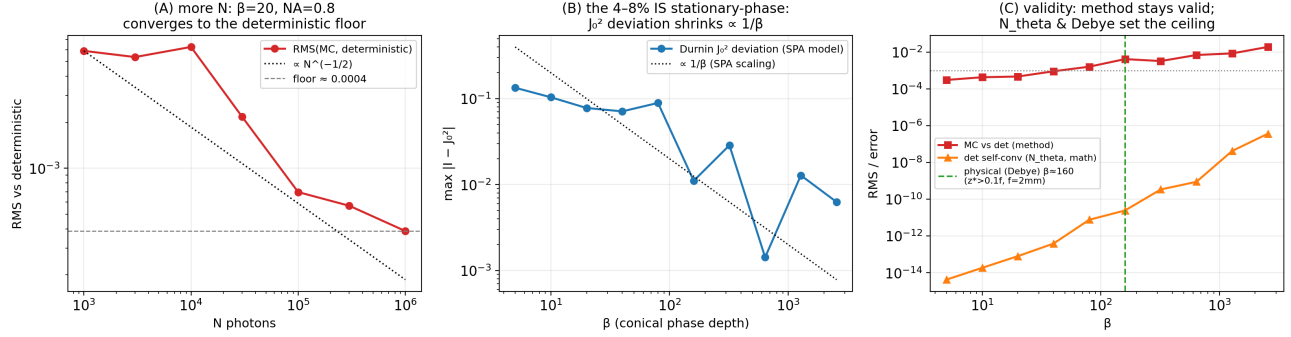
Verification — push N and β to the validity boundary (axicon_validity.png)

The 4–8% above is the stationary-phase *model*, not the method:

- **More photons** ($\beta=20$, $NA=0.8$): RMS(MC–deterministic) falls $\propto 1/\sqrt{N}$ to a floor $\approx 4 \times 10^{-4}$ at $N = 10^6$.
- **Push β** ($NA=0.3$, scalar regime, $\theta^*=0.7\alpha$): the J_0^2 deviation shrinks $0.134 \rightarrow 0.078$ ($\beta=20$) $\rightarrow 0.006$ ($\beta=2560$), $\approx 1/\beta$ — confirming SPA; MC vs deterministic stays ≤ 0.019 at every β (the method is bias-free).

- **Validity ceilings:** $N_\theta=3001$ resolves the pupil phase to $\beta > 2560$ (mathematical); for a real objective ($f \approx 2$ mm) the focal segment $z^* \propto \beta$ leaves the focal region (Debye) at $\beta \approx 160$ — the physical limit binds first.

Axicon: more photons, then push β to the validity boundaries



(A) more $N \rightarrow$ deterministic floor; (B) J_0^2 deviation $\propto 1/\beta$ (it IS stationary-phase); (C) method stays valid, N_θ & Debye set the ceiling.